

Naresh Enugala

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Professional Summary:

- 3 years 4 months of experience in VLSI industry as a Physical design engineer.
- Worked in 10 nm, 14 nm and 28nm technology nodes.
- Expert in Physical Design flow: Initial Design checks, Floorplan, Placement, Clock Tree Synthesis, Routing and Timing Closure.
- Hands on experience in EDA tools like ICC2 and Primetime and Star-RC.
- Good knowledge on TCL and shell scripting.
- Good Knowledge in timing fixes like DRVs, Setup, Hold, Crosstalk etc.

Skill set:

- **Tools** : ICC2, Star-RC, Primetime
- **Nodes** : 10nm, 14nm, 28nm
- **Language** : TCL, shell
- **Technical Skills** : Physical Design, Static Timing Analysis, Sign off, ECO.

PROFESSIONAL EXPERIENCE

Company: Scaledge Physical Design Engineer Nov 2022- Oct 2023

Company: MiraFra Technology Physical Design Engineer Jul 2022- Oct 2022

Company: ChipOnTime Technology Physical Design Engineer Jun 2020- Jul 2022

Education Details:

- B. Tech in Electronics and Communication Engineering from CMR Engineering College in 2016 with 63.6%.

Project Details:

Project 1:

Technology Node : 10 nm

Design Statistics : Primary Clock Max Frequency 1GHz, 2M instances, 36 Macros.

Tools Used : ICC2, Star-RC, Primetime
Project : GF10nm
Client : Global Foundries

Responsibilities:

- Performed complete Physical design activity starting from Floor planning, Placement, CTS, Routing.
- Placed macros based on communication to avoid Criss cross connections.
- Done Multiple Floor plan iterations to balance between Routing and Timing issues.
- I have applied keep out margin to avoid the congestion due to high pin density cells like AOI, OAI cells.
- Apply Group paths and bounds to meeting timing at placement stage.
- Achieved skew and latency targets by putting parallel runs regards different drive strengths.
- Resolved the timing issues at Placement stage by creating Bounds where the Module placement was not good
- Usage of Partial blockages to prevent clustering of cells near memories to avoid high Local placement density & routing congestion.
- Performed the timing Closure using prime time and generated the ECOs using Tweaker.

Project 2:

Technology Node : 14 nm
Design Statistics : Primary Clock Max Frequency 600MHz, 1.5M instances, 24Macros.
Tools Used : ICC2
Project : GF14nm
Client : Global Foundries

Responsibilities:

- Performed complete physical design activity starting from Sanity Checks, Floor planning, placement, CTS and routing.
- Placed macros by using Data flow Diagram Performed different congestion experiments.
- Usage of Partial blockages to prevent clustering of cells to avoid cell density.
- Improving timing during placement, minimizing skew and insertion delay during CTS.
- Built the clock tree properly with NDRs. Fixed clock insertion delays and skewing issues in CTS.
- Fixed the setup, hold and DRVs violations at block level.

Project 3:

Technology Node : 28 nm

Design Statistics : Primary Clock Max Frequency 525 MHz, 2M instances, 38 Macros.
Tools Used : ICC2
Client : Global Foundries

Responsibilities:

- The tasks handled were Floor planning, Place & Route of the design, bring the block to timing closure.
- Took us more than couple of iterations to arrive at best floor plan.
- This block is congestion critical.
- I have taken care of both timing and congestion results at each stage in PNR Flow.